



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 20th April 2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई -400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु -560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 19.04.2024.

महोदय/Dear Sir,

आईईजीसी-2010 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 19-अप्रैल-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 19th April 2024, is available at the NLDC website.

धन्यवाद,

Report for previous day

Date of Reporting: 20-Apr-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	59035	66236	50476	28982	2480	207209
Peak Shortage (MW)	0	0	0	0	0	0
Energy Met (MU)	1261	1628	1288	635	51	4864
Hydro Gen (MU)	165	49	47	31	14	307
Wind Gen (MU)	61	115	52	-	-	228
Solar Gen (MU)*	169.26	90.39	120.18	3.98	0.65	384
Energy Shortage (MU)	0.19	0.00	0.00	2.99	0.00	3.18
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	59172	73980	60161	29739	2754	214645
Time Of Maximum Demand Met	22:23	15:14	14:52	23:30	18:51	22:42

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.050	0.00	0.19	4.48	4.66	71.65	23.69

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	8808	0	155.3	55.1	-1.3	200	0.00
	Haryana	8687	0	166.7	104.7	-0.8	189	0.00
	Rajasthan	12874	0	260.6	28.1	-7.9	340	0.00
	Delhi	5036	0	104.3	94.0	1.3	248	0.00
	UP	23032	0	448.2	203.5	-5.1	956	0.00
	Uttarakhand	1869	0	38.2	17.6	-2.0	210	0.00
	HP	1610	0	30.8	15.5	-0.4	63	0.19
	J&K(UT) & Ladakh(UT)	2415	0	48.5	34.8	-0.1	310	0.00
	Chandigarh	236	0	4.8	4.8	-0.1	22	0.00
WR	Railways_NR ISTS	208	0	4.0	3.8	0.2	70	0.00
	Chhattisgarh	6160	0	139.4	79.2	-0.9	219	0.00
	Gujarat	23916	0	491.8	175.6	-3.6	456	0.00
	MP	12820	0	279.0	145.3	-3.0	450	0.00
	Maharashtra	28446	0	640.5	229.8	-3.4	746	0.00
	Goa	803	0	17.2	12.1	4.8	60	0.00
	DNHDDPDCL	1270	0	29.4	29.5	-0.1	38	0.00
	AMNSIL	806	0	18.3	5.6	0.1	276	0.00
	BALCO	521	0	12.5	12.3	0.2	87	0.00
SR	Andhra Pradesh	13319	0	253.6	93.5	0.8	1009	0.00
	Telangana	11343	0	239.2	124.0	-0.4	513	0.00
	Karnataka	15656	0	298.6	148.2	2.5	999	0.00
	Kerala	5422	0	108.4	86.2	1.5	276	0.00
	Tamil Nadu	18334	0	379.6	242.7	-11.5	809	0.00
	Puducherry	449	0	8.8	8.6	-0.4	25	0.00
ER	Bihar	6663	0	130.1	120.7	-1.5	422	1.30
	DVC	3172	0	72.6	-43.3	0.6	271	0.00
	Jharkhand	2083	0	38.7	34.6	-3.8	343	1.69
	Odisha	6701	0	148.5	55.7	0.4	467	0.00
	West Bengal	11887	0	244.0	98.7	-1.3	487	0.00
	Sikkim	70	0	1.3	1.3	-0.1	23	0.00
	Railways_ER ISTS	11	0	0.1	0.1	0.0	4	0.00
NER	Arunachal Pradesh	128	0	2.0	2.7	-0.9	49	0.00
	Assam	1659	0	32.0	25.6	0.0	91	0.00
	Manipur	200	0	2.8	2.9	-0.1	23	0.00
	Meghalaya	345	0	5.6	4.8	-0.2	82	0.00
	Mizoram	113	0	1.6	1.8	-0.4	4	0.00
	Nagaland	129	0	2.5	2.5	-0.1	2	0.00
	Tripura	288	0	4.6	4.5	0.1	82	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	0.9	-9.7	-24.1	-23.9
Day Peak (MW)	109.0	-741.6	-1060.0	-1314.7

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	110.0	-329.0	251.0	-35.7	3.6	0.0
Actual(MU)	92.5	-334.7	271.5	-31.1	3.2	1.3
O/D/U/D(MU)	-17.5	-5.7	20.5	4.5	-0.4	1.3

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	3035	4229	2968	3766	155	14152	42
State Sector	7401	5947	3918	2464	150	19879	58
Total	10435	10176	6886	6229	305	34030	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	771	1675	703	719	14	3882	74
Lignite	20	18	67	0	0	106	2
Hydro	165	49	47	31	14	307	6
Nuclear	25	64	72	0	0	161	3
Gas, Naptha & Diesel	37	87	6	0	25	155	3
RES (Wind, Solar, Biomass & Others)	241	209	204	6	1	660	13
Total	1259	2103	1099	756	54	5271	100

Share of RES in total generation (%)	19.15	9.96	18.54	0.75	1.24	12.53
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	34.25	15.34	29.39	4.85	27.45	21.40

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.051
Based on State Max Demands	1.106

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	208306	14:59	0
Non-Solar hr	214645	22:42	6

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 07:00 to 17:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 20-Apr-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	0	0.0	0.0	0.0
2	HVDC	PUSAULI B/B	-	0	48	0.0	1.2	-1.2
3	765 kV	GAYA-VARANASI	2	839	523	2.9	0.0	2.9
4	765 kV	SASARAM-FATEHPUR	1	212	347	0.0	2.6	-2.6
5	765 kV	GAYA-BALIA	1	131	497	0.0	6.1	-6.1
6	400 kV	PUSAULI-VARANASI	1	0	108	0.0	1.4	-1.4
7	400 kV	PUSAULI -ALLAHABAD	1	61	32	0.2	0.0	0.2
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	821	319	4.3	0.0	4.3
9	400 kV	PATNA-BALIA	2	459	737	0.0	6.2	-6.2
10	400 kV	NAUBATPUR-BALIA	2	0	228	0.0	1.9	-1.9
11	400 kV	BIHARSHARIFF-BALIA	2	569	59	5.9	0.0	5.9
12	400 kV	MOTIHARI-GORAKHPUR	2	402	370	0.0	1.3	-1.3
13	400 kV	BIHARSHARIFF-VARANASI	2	242	110	1.2	0.0	1.2
14	220 kV	SAHUPURI-KARAMNASA	1	47	98	0.0	1.5	-1.5
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	30	0	0.7	0.0	0.7
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						15.2	22.1	-6.9
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	2146	0	35.4	0.0	35.4
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	1233	106	13.1	0.0	13.1
3	765 kV	JHARSUGUDA-DURG	2	0	701	0.0	11.0	-11.0
4	400 kV	JHARSUGUDA-RAIGARH	4	0	431	0.0	5.6	-5.6
5	400 kV	RANCHI-SIPAT	2	241	104	0.9	0.0	0.9
6	220 kV	BUDHIPADAR-RAIGARH	1	32	123	0.0	1.0	-1.0
7	220 kV	BUDHIPADAR-KORBA	2	0	2	0.0	0.0	0.0
ER-WR						49.4	17.6	31.8
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	573	0.0	12.0	-12.0
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	1494	0.0	36.1	-36.1
3	765 kV	ANGUL-SRIKAKULAM	2	0	2930	0.0	56.5	-56.5
4	400 kV	TALCHER-I/C	2	420	474	5.5	0.0	5.5
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	104.6	-104.6
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	327	130	2.3	0.3	2.1
2	400 kV	ALIPURDUAR-BONGAIGAON	2	595	149	4.2	0.0	4.2
3	220 kV	ALIPURDUAR-SALAKATI	2	79	40	0.4	0.0	0.4
ER-NER						6.9	0.3	6.7
Import/Export of <null> (With <null>)								
No Records Found								
NER-NR						0.0	0.0	0.0
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	2011	0.0	48.0	-48.0
2	HVDC	VINDHYACHAL B/B	-	492	0	12.2	0.0	12.2
3	HVDC	MUNDRA-MOHINDERGARH	2	0	1449	0.0	27.4	-27.4
4	765 kV	GWALIOR-AGRA	2	583	1787	1.9	20.0	-18.2
5	765 kV	GWALIOR-PHAGI	2	1894	1060	11.4	7.3	4.2
6	765 kV	JABALPUR-ORAI	2	574	691	0.0	8.1	-8.1
7	765 kV	GWALIOR-ORAI	1	553	0	5.0	0.0	5.0
8	765 kV	SATNA-ORAI	1	0	1023	0.0	17.9	-17.9
9	765 kV	BANASKANTHA-CHITORGARH	2	1464	266	12.0	0.5	11.5
10	765 kV	VINDHYACHAL-VARANASI	2	0	2956	0.0	46.4	-46.4
11	400 kV	ZERDA-KANKROLI	1	410	0	4.4	0.0	4.4
12	400 kV	ZERDA -BHINMAL	1	409	0	4.4	0.0	4.4
13	400 kV	VINDHYACHAL -RIHAND	1	967	0	21.0	0.0	21.0
14	400 kV	RAPP-SHUJALPUR	2	975	185	7.4	0.7	6.8
15	400 kV	NEEMUCH-Chittorgarh	2	1120	119	8.5	0.3	8.2
16	220 kV	BHANPURA-RANPUR	1	0	132	0.0	2.3	-2.3
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.0	-2.0
18	220 kV	MEHGAON-AURAIYA	1	112	1	1.2	0.0	1.2
19	220 kV	MALANPUR-AURAIYA	1	109	0	1.0	0.0	1.0
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						90.5	180.8	-90.3
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1011	0.0	20.8	-20.8
2	HVDC	RAIGARH-PUGALUR	2	0	6323	0.0	80.7	-80.7
3	765 kV	SOLAPUR-RAICHUR	2	0	1665	0.0	24.8	-24.8
4	765 kV	WARDHA-NIZAMABAD	2	0	2656	0.0	46.4	-46.4
5	765 kV	WARORA-WARANGAL(NEW)	2	0	3028	0.0	54.8	-54.8
6	400 kV	KOLHAPUR-KUDGI	2	709	0	10.9	0.0	10.9
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	0	98	1.8	0.0	1.8
9	220 kV	XELDEM-AMBEWADI	1	0	129	2.7	0.0	2.7
WR-SR						15.4	227.4	-212.0

INTERNATIONAL EXCHANGES						Import(+ve)/Export(-ve)
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	268	-23	109	2.61
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	103	-73	30	0.72
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	-178	-114	-141	-3.38
	NER	132kV GELEPHU-SALAKATI	10	0	3	0.06
	NER	132kV MOTANGA-RANGIA	47	30	38	0.91
NEPAL	NR	NEPAL IMPORT (FROM UP)	-77	0	-27	-0.65
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	-77	0	-39	-0.93
	ER	NEPAL IMPORT (FROM BIHAR)	-216	-43	-69	-1.66
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	-526	-1	-269	-6.46
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-946	-790	-904	-21.69
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1315	-810	-995	-23.89
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-114	0	-100	-2.40

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	1.10	0.00	0.00	0.00	0.00	0.00	1.10
Nepal	0.32	0.00	6.37	0.00	0.00	0.00	0.00	0.00	6.69
Bangladesh	21.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.73
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	22.05	0.00	7.47	0.00	0.00	0.00	0.00	0.00	29.52

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	1.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.35
Nepal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	1.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.35

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	1.35	0.00	-1.10	0.00	0.00	0.00	0.00	0.00	0.25
Nepal	-0.32	0.00	-6.37	0.00	0.00	0.00	0.00	0.00	-6.69
Bangladesh	-21.73	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-21.73
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	-20.70	0.00	-7.47	0.00	0.00	0.00	0.00	0.00	-28.17